Tuan Anh Pham Ph.D.

tuan.pham@ed.ac.uk
https://webhomes.maths.ed.ac.uk/~tpham/

Education

University of Edinburgh — Edinburgh, United Kingdom
Ph.D. in Algebra and Number Theory — 2022 - 2026 *(expected)*

- **Thesis title:** Primitive Ideals of the Enveloping Algebra of the Witt Algebra
- Advisor: Prof. Susan J. Sierra

Jacobs University Bremen (Constructor University) — Bremen, Germany
BSc in Mathematics and Computer Science — 2019 - 2022

- GPA (German-scale): 1.0 (highest)
- President's List in Mathematics and Computer Science department from 2019-22
- **Bachelor thesis:** On the problem of computing the Ext-groups in a category of tensor modules over a Mackley Lie algebra
- Advisor: Prof. Ivan Penkov

Publications and Preprints

1. **Tuan Anh Pham**, James Timmins. The kernel and image of orbit homomorphisms for the Witt algebra. *arXiv:2510.00756*.
2. Lucas Buzaglo, Xiao He, **Tuan Anh Pham**, Haijun Tan, Girish S Vishwa, Kaiming Zhao. On the boundary Carrollian conformal algebra. *arXiv:2508.21603*.
3. **Tuan Anh Pham**. The orbit method for the Virasoro algebra. *arXiv:2504.14670*.
4. Tina Boroukhian, Kritkorn Supyen, Christopher William Mclaughlan, Atit Bashyal, **Tuan Anh Pham**, Hendro Wicaksono. Semantic middleware for demand response systems: Enhancing data interoperability in green electricity management for manufacturing. *Computers in Industry, 2025*.
5. **Tuan Anh Pham**, Vikas Garg. What do Graph Neural Networks learn? Insights from Tropical Geometry. *NeurIPS 2024*.

Work Experience

Aalto University | Helsinki, Finland — 2022.06-2022.09

- Research assistant on the project Graph Neural Network with Prof. Vikas Garg. Our paper was published later in NeurIPS 2024.

Jacobs University Bremen | Bremen, Germany — 2020.12 - 2022.06

- Research assistant in Delfine group with Prof. Hendro Wicaksono on formalizing and developing machine learning models for ontologies. Our paper was published later in Computers in Industry, 2025.

Research Assistant in Alfred Wegener Institute | Bremen, Germany — 2021.08 - 2021.09

Research Assistant in Ruhr University | Bochum, Germany — 2021.06 - 2021.07

Teaching Assistant | Bremen, Germany — 2020.09 – 2020.12

Research Talks

- **Primitive and two-sided ideals of the Witt algebra**, Algebra, Geometry and Number Theory Seminar at Bath, Nov 2025
- **The orbit method and orbit homomorphisms for the Witt algebra**, Séminaire d'algèbre et de géométrie, Laboratoire de mathématiques Nicolas Oresme (LMNO), Caen, Oct 2025
- **The orbit method for the Virasoro algebra**, X International Workshop on Non-Associative Algebras in Dijon, France, Sep 2025
- **The orbit method for the Virasoro algebra**, European Non-Associative Algebra Seminar, online, July 2025
- **The orbit method for the Virasoro algebra**, Hopf25 conference, Brussels, April 2025
- **Prime and primitive ideals in noncommutative rings**, Hodge Club, Edinburgh, April 2025
- **The orbit method for the Virasoro algebra**, Hodge Algebra seminar, Edinburgh, April 2025

- **The orbit method for the Witt algebra**, International Conference on Representations of Algebras (ICRA 21), Shanghai, August 2024
- **The orbit method for the Witt algebra**, Oxford Junior Algebra and Representation Theory Seminar, Oxford May 2024
- **Local representations of the Witt algebra**, GEARS seminar, Glasgow, March 2024
- **Local representations of the Witt algebra**, Hodge Club seminar, Edinburgh, Feb 2024

Expository Talks

- **Prime ideals of quantum algebras**, Hodge Club seminar, Edinburgh, Oct 2025
- **Computations for Lie algebras of formal vector fields. General results**, Reading group on Cohomology of infinite-dimensional Lie algebras, Edinburgh, Nov 2024
- **Computations for Lie algebras of smooth vector fields**, Reading group on Cohomology of infinite-dimensional Lie algebras, Edinburgh, Oct 2024
- **Poisson Varieties and Symplectic Leaves**, Hodge Seminar pretalk, Edinburgh, Feb 2024
- **Functions and subalgebras on the Witt algebra**, GlaMS Example Showcase, Edinburgh, Feb 2023

Awards and Grants

- RepNet travel support for Dijon 2025
- RepNet travel support for a research visit with Prof. Stéphane Launois, Université de Caen Normandie, Sep 2025
- Laura Wisewell travel scholarship. £700 for ICRA2021
- ERAMUS+ scholarship for internship semester in Helsinki 2022
- Jacobs University merit-based Full-tuition "Diversity" Scholarship 2020-22
- ERAMUS+ scholarship for exchange semester at University of Aberdeen 2021
- Jacob University President's List from 2019-22
- Sparkassen Mathematics Scholarship for high academic performance 2019-22

Teaching Experience

- Advanced Mathematical Economics, Edinburgh, Fall 2025
- Count Me In programme (Research Mentor and Tutor), Edinburgh, Summer 2025
- AGQ Advanced Research Computing Spring 2025
- Proofs and Problem Solving, Edinburgh, Spring 2023, Spring 2024, Spring 2025
- Introduction to Number Theory, Edinburgh, Spring 2024, Spring 2025
- Commutative Algebra, Edinburgh, Fall 2024
- Group Theory, Edinburgh, Fall 2024
- Honours Algebra, Edinburgh, Spring 2024
- Honours Analysis skills, Edinburgh, Fall 2023
- Combinatorics and Graph Theory, Edinburgh, Fall 2023
- Facets of Mathematics, Edinburgh, Fall 2023

Organization Experience

- Postgraduate Research Committee in the School of Mathematics, Edinburgh, 2025-
- Organizer, GEARS seminar (Glasgow Edinburgh Algebra Research Student seminar), 2024-2025, with Oli Jones and Parth Shimpi
- Organizer, Reading group on Cohomology of Infinite-Dimensional Lie Algebras, 2024, with Susan Sierra
- Organizer, Hodge Club, the seminar for Hodge Institute graduate students and postdocs, 2024-2025, with Yan Yau Cheng and Danil Kozenikov
- Reviewer for zbMATH Open

Profficiencies

- **Programming languages:** Sage, Python, Lean, C/C++, Machine Learning models
- **Languages:** Vietnamese (native), English (professional), German (intermediate)